Syntax:

Function Header

[return data type] Function_name([parameters/arguments])

{

Statements;

[return value;]

Function Body

}

Here return data type indicates the type of value that the function is returning to main() / called function. It is depended on the return value.

If the sub program is having return statement then it is called function and without return value it is called procedure.

The **default return value of a function is integer**. Use the keyword **void** [nothing] when the function is not having any return value. Otherwise integer will become the return value.

A function may have any number of return statements. But only one return statement is executed at a time. The statements after the return statement are not executed and gives compile time warning.

Return data type should be matched with return value data type.

Function name is used to identify a function and it should not be predefined.

Arguments / parameters used to carry the values from function calling area to function definition area. They are optional and of two types.

1. **Actual** arguments/parameters.
2. **Formal** arguments/parameters.

The arguments we are sending in function calling at main() are called actual parameters and the parameters we are using in function definition to receive the actual argument values is called formal parameters.

Actual and formal parameter names may be same or different.

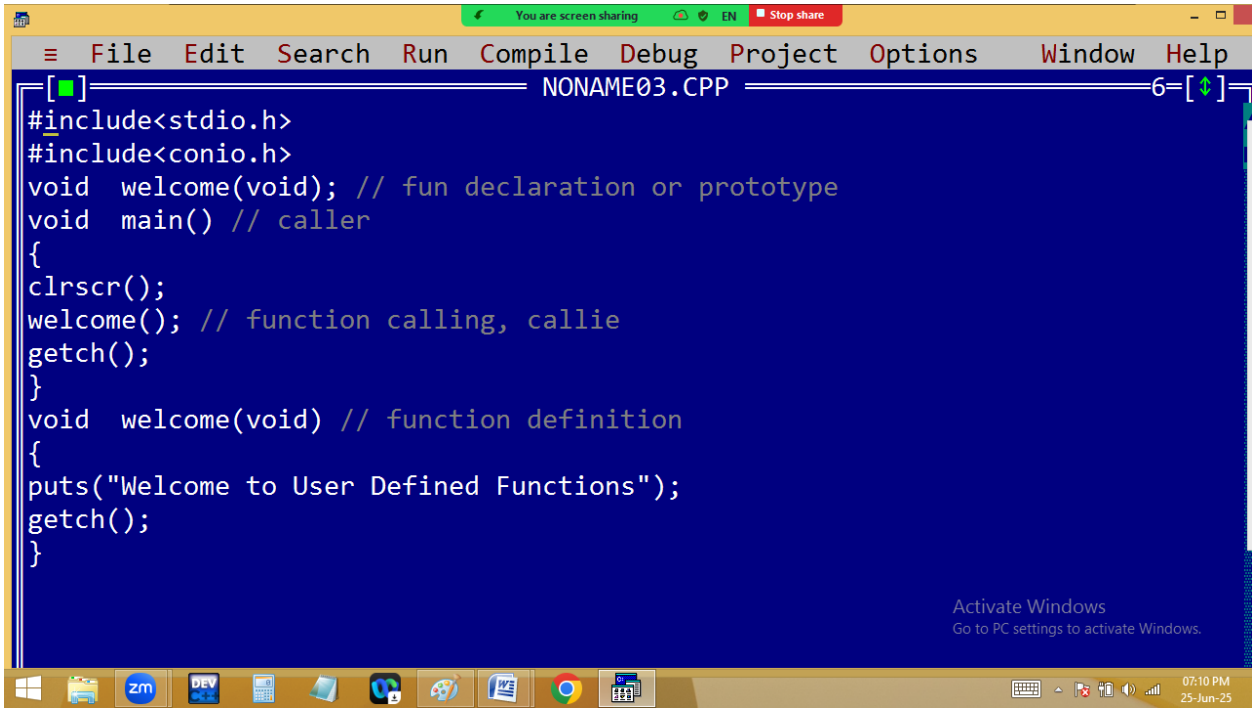
Difference between arguments and parameters is "arguments are nothing but different vehicles which are parked in same parking area and the parking area is the parameters. i.e. arguments are always changed but parameters are fixed.

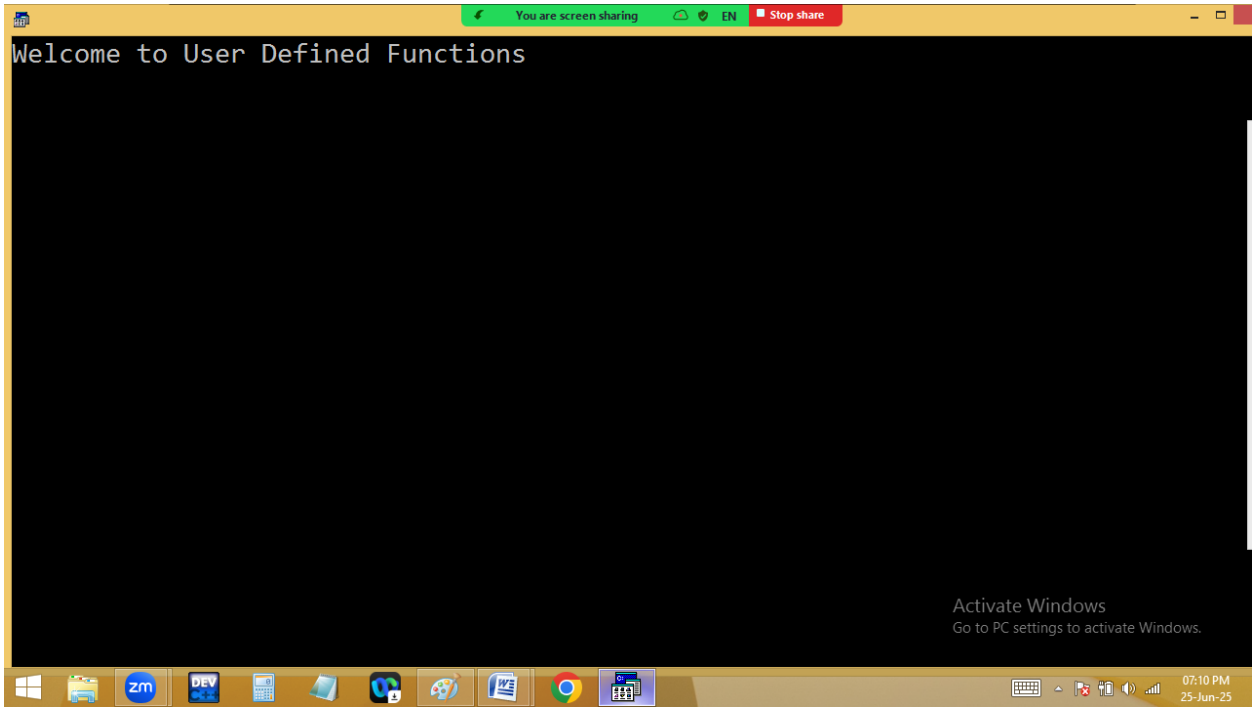
The calling function arguments should be matched with definition parameters list in quantity, data type and order.

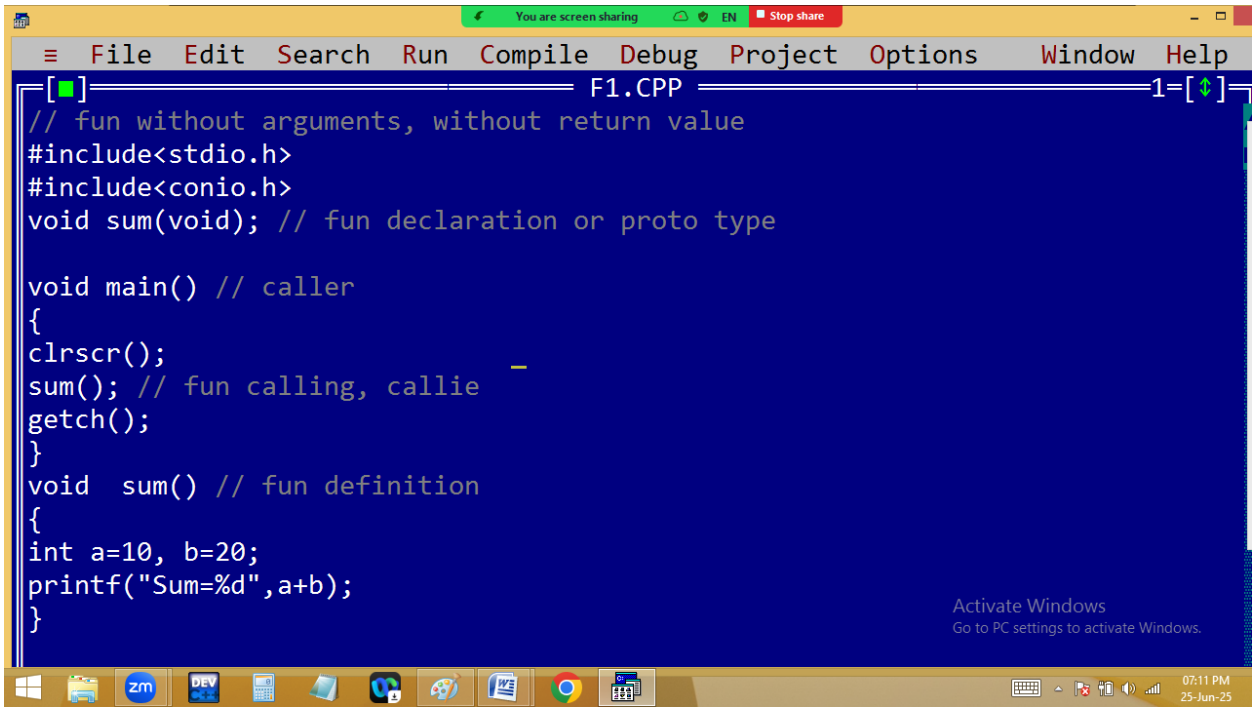
Based on arguments and return values, functions are divided into 4 types.

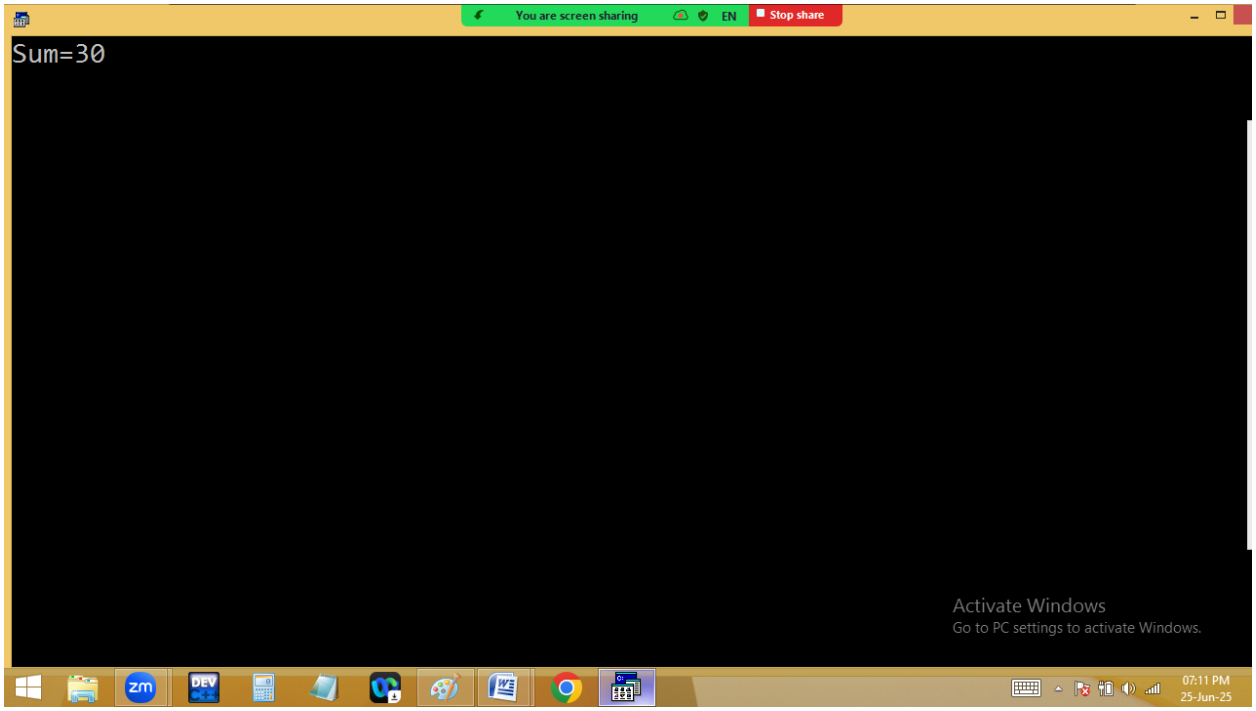
1. Function without arguments, without return values.
2. Function with arguments and with return values.
3. Function with arguments and without return values.
4. Function without arguments and with return values.

Function without arguments, without return values.

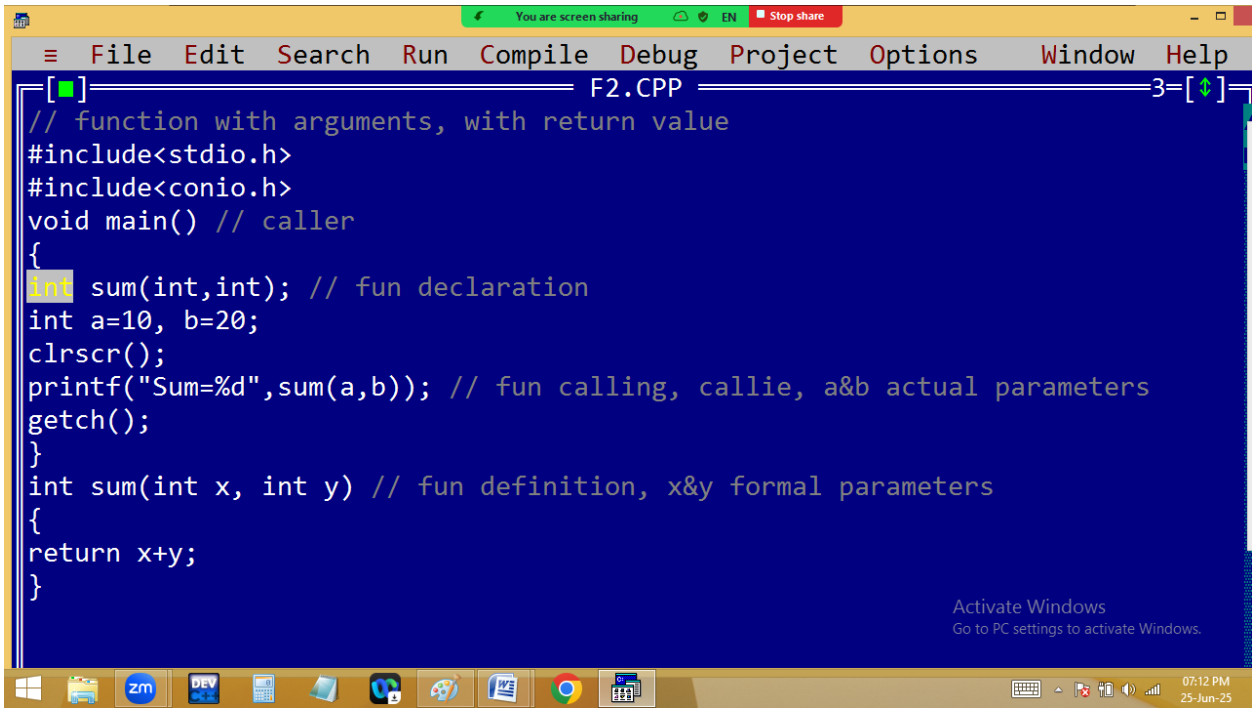

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NONAME03.CPP
#include<stdio.h>
#include<conio.h>
void welcome(void); // fun declaration or prototype
void main() // caller
{
clrscr();
welcome(); // function calling, callie
getch();
}
void welcome(void) // function definition
{
puts("Welcome to User Defined Functions");
getch();
}
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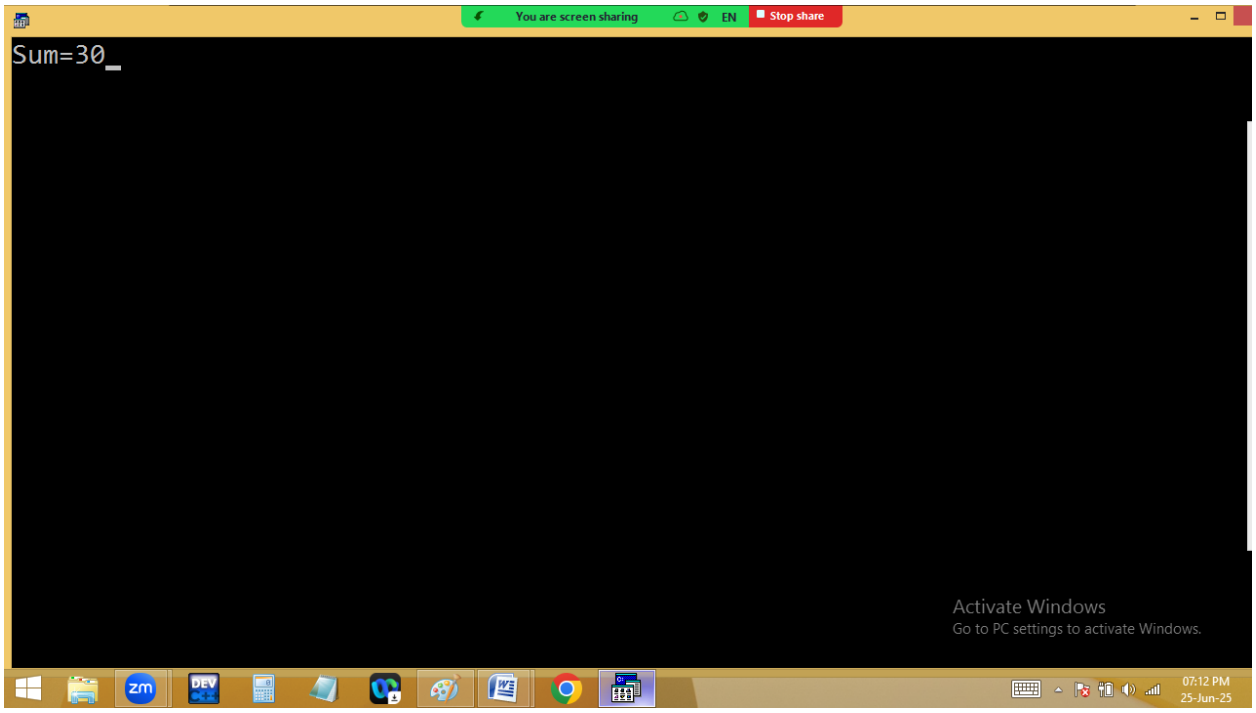

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Welcome to User Defined Functions
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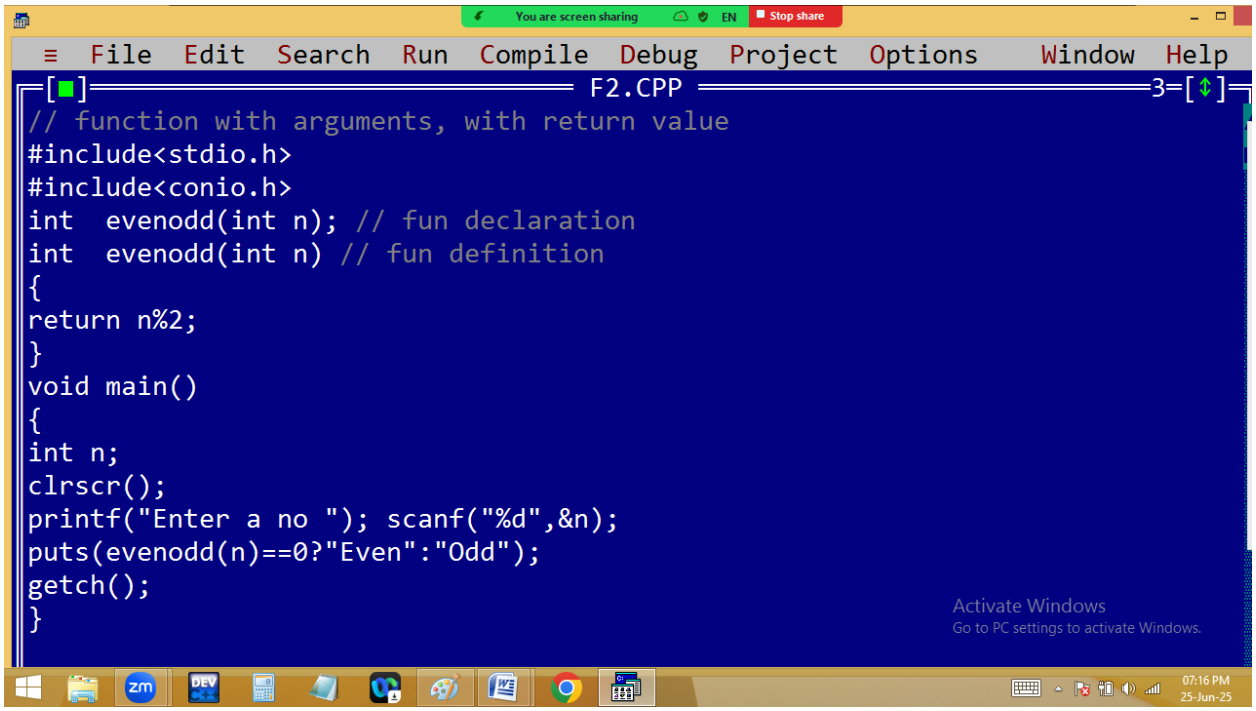

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F1.CPP
// fun without arguments, without return value
#include<stdio.h>
#include<conio.h>
void sum(void); // fun declaration or proto type
void main() // caller
{
clrscr();
sum(); // fun calling, callie
getch();
}
void sum() // fun definition
{
int a=10, b=20;
printf("Sum=%d",a+b);
}
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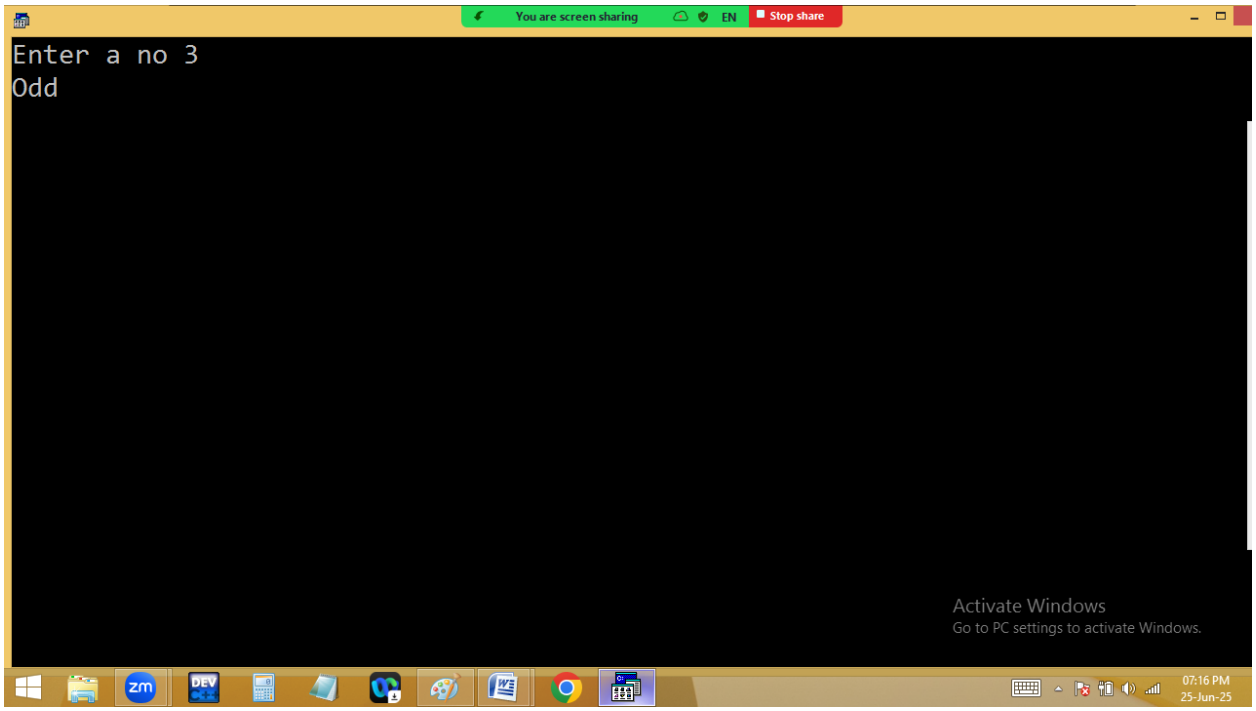

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Sum=30
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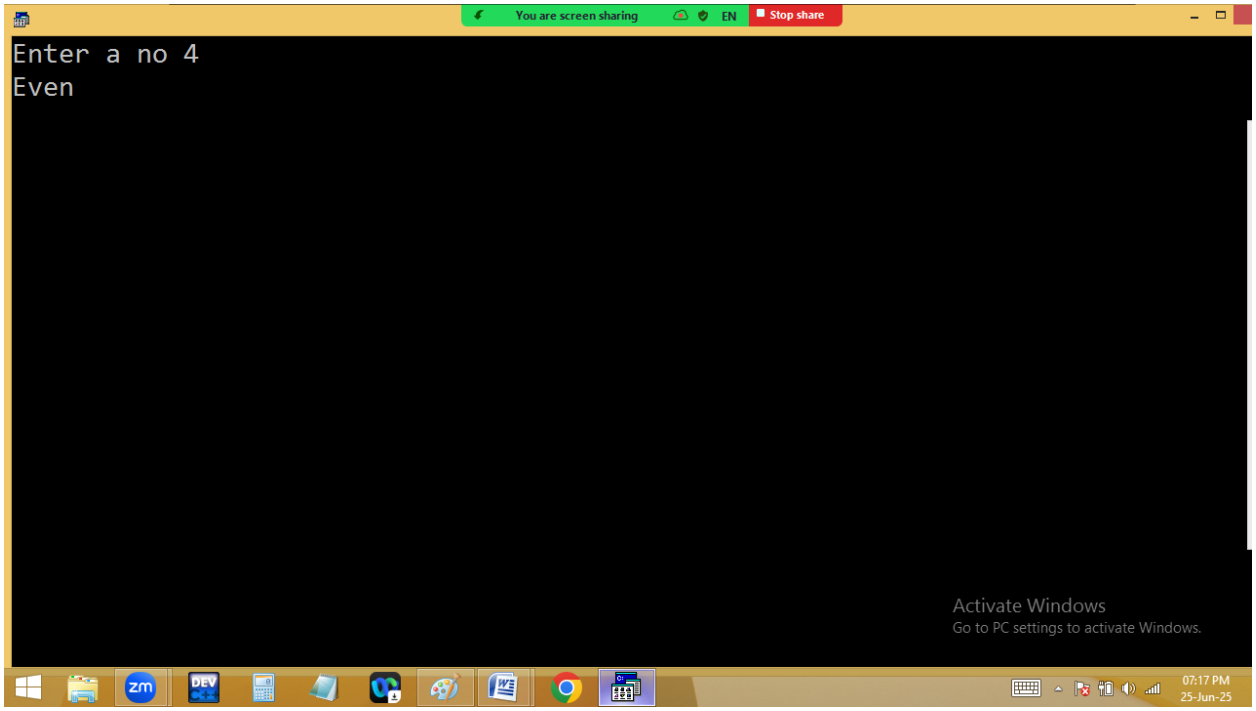
Function with arguments and with return value:



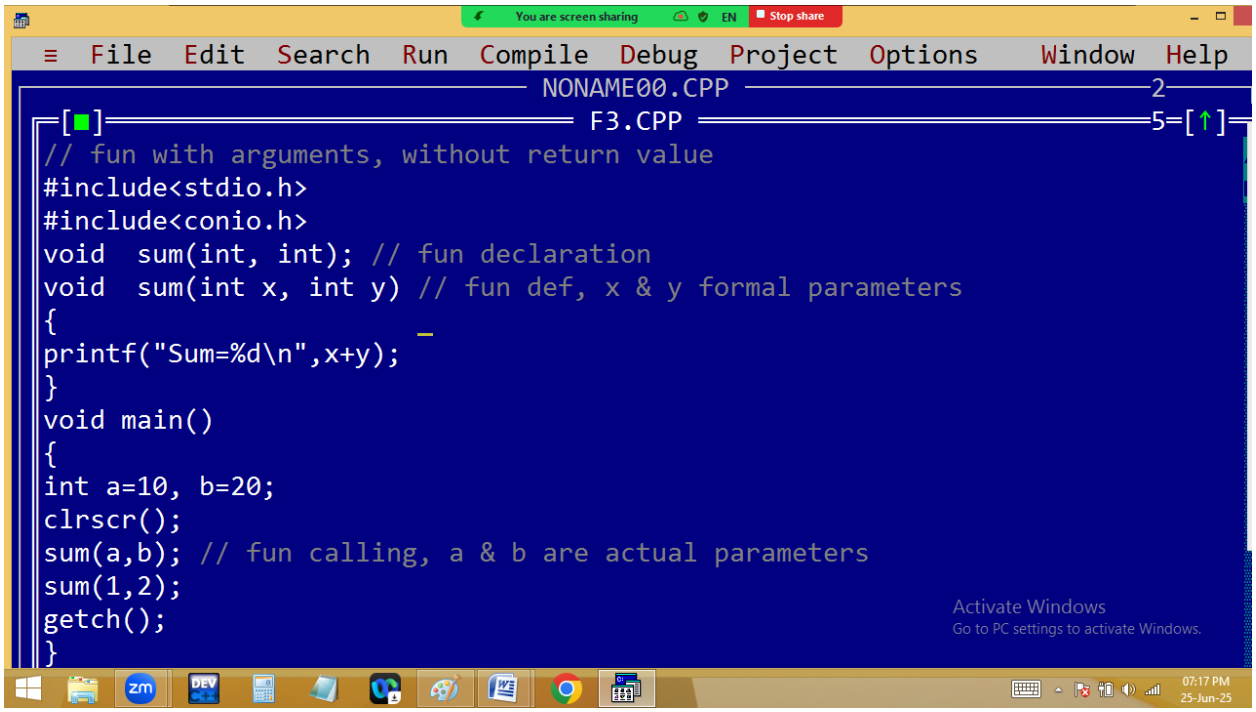


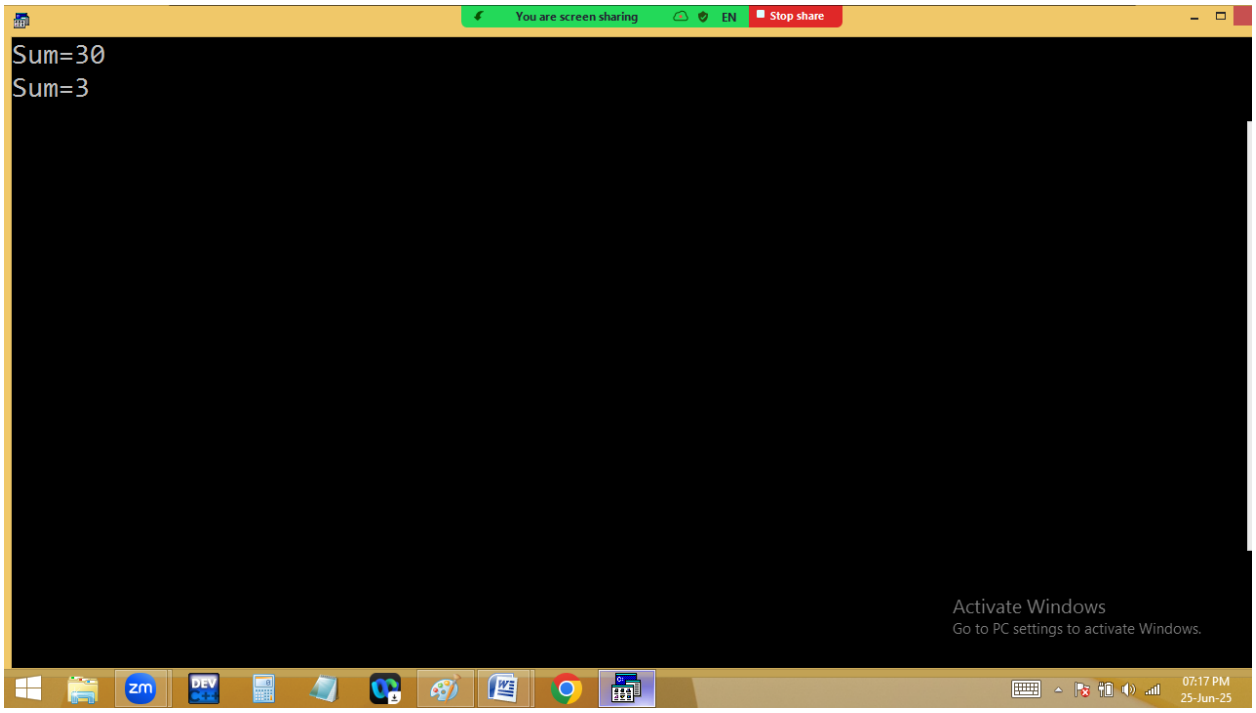

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F2.CPP
// function with arguments, with return value
#include<stdio.h>
#include<conio.h>
int evenodd(int n); // fun declaration
int evenodd(int n) // fun definition
{
return n%2;
}
void main()
{
int n;
clrscr();
printf("Enter a no "); scanf("%d",&n);
puts(evenodd(n)==0?"Even":"Odd");
getch();
}
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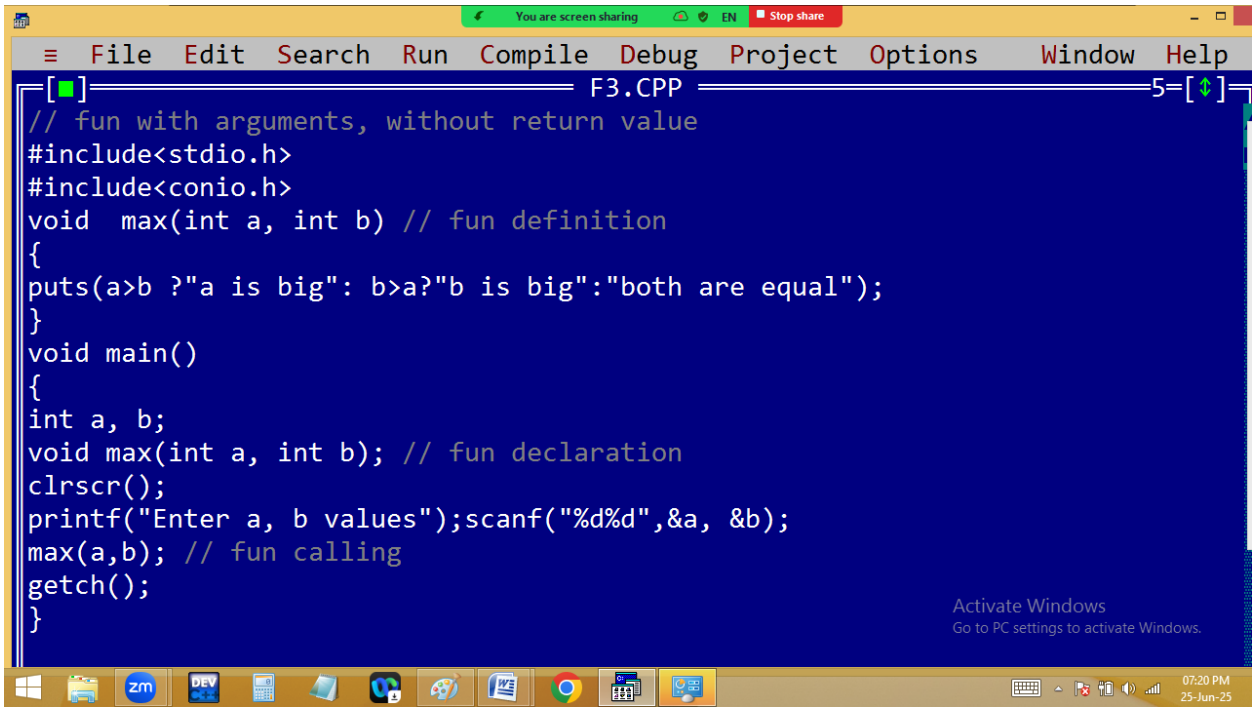

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Enter a no 3
Odd
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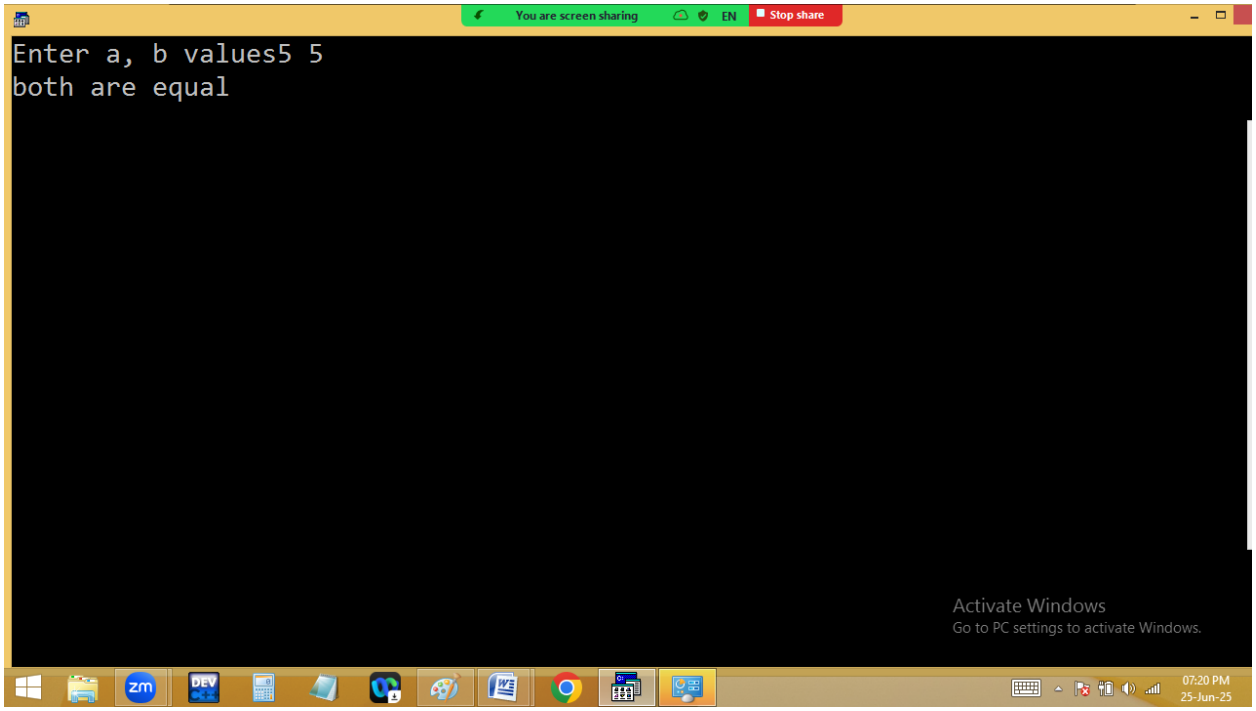

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Enter a no 4
Even
Activate Windows
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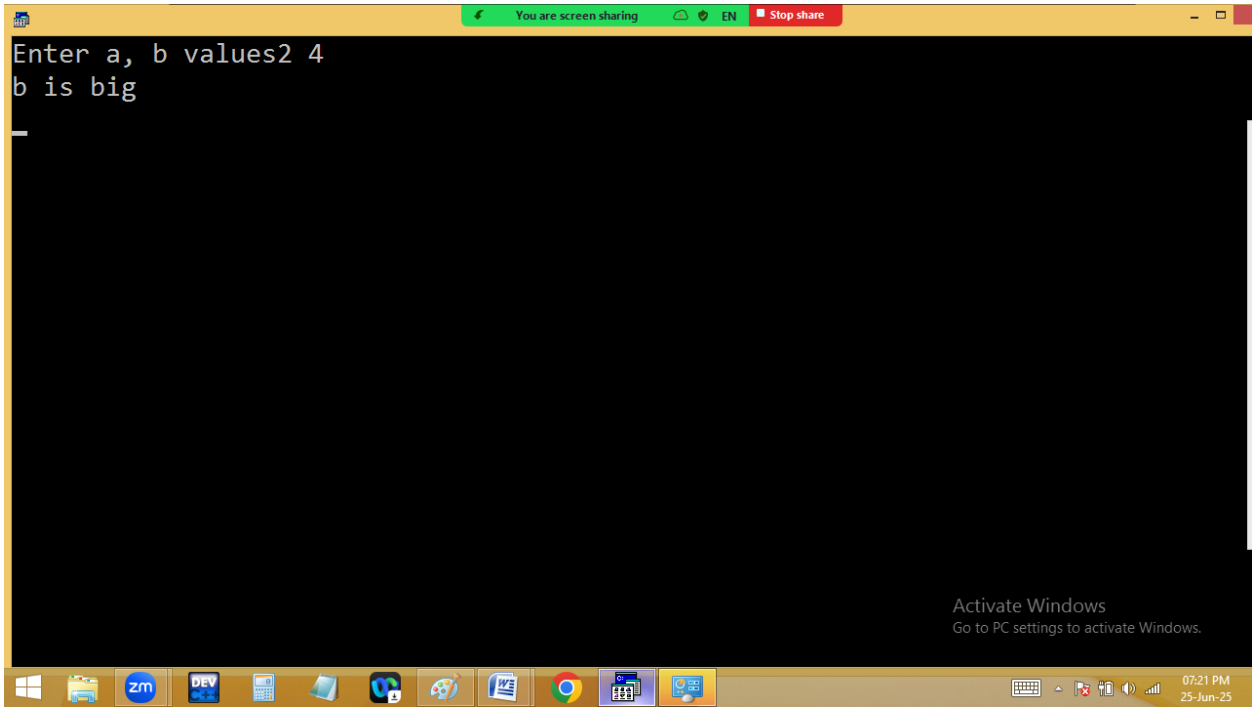
Function with arguments, without return value:



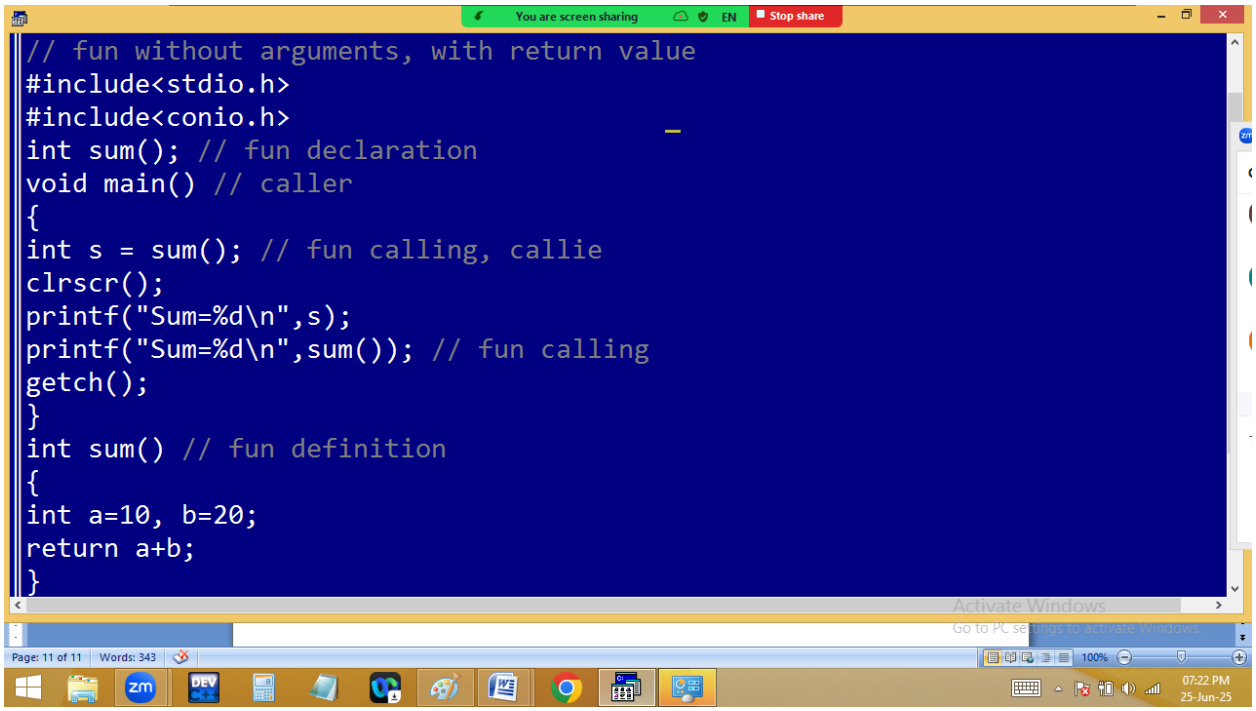


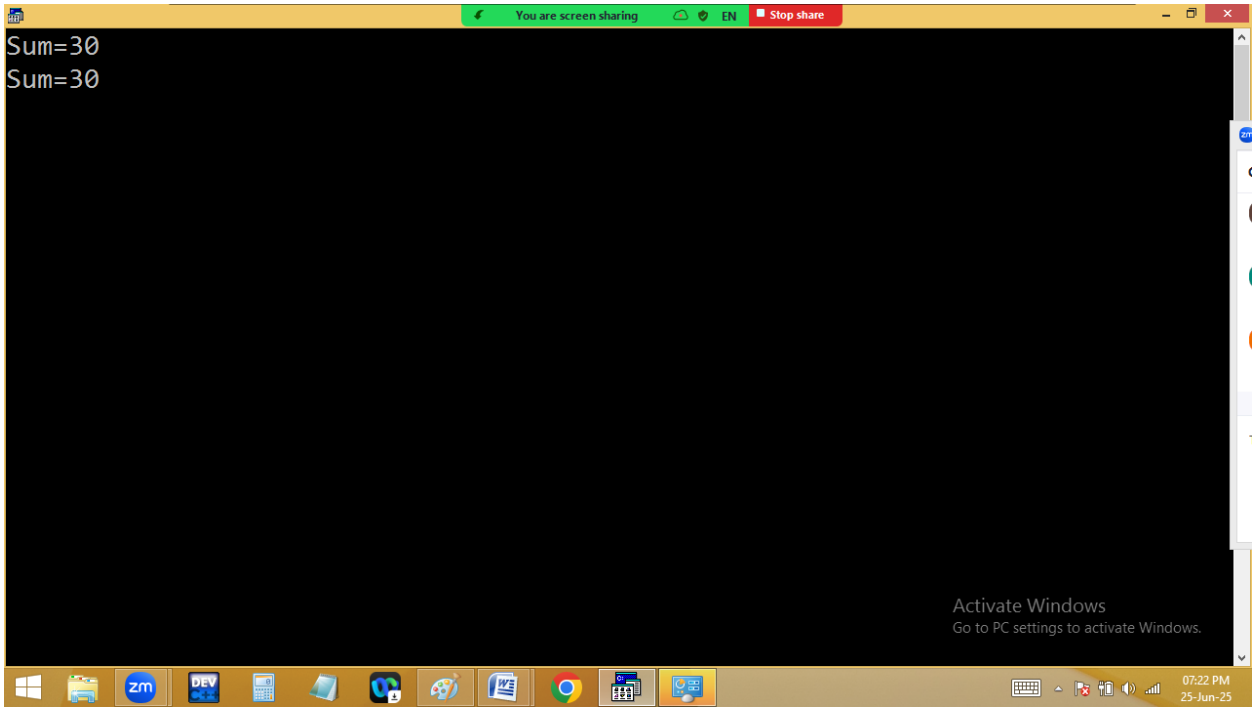

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F3.CPP
// fun with arguments, without return value
#include<stdio.h>
#include<conio.h>
void max(int a, int b) // fun definition
{
puts(a>b ?"a is big": b>a?"b is big":"both are equal");
}
void main()
{
int a, b;
void max(int a, int b); // fun declaration
clrscr();
printf("Enter a, b values");scanf("%d%d",&a, &b);
max(a,b); // fun calling
getch();
}
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Enter a, b values5 5
both are equal
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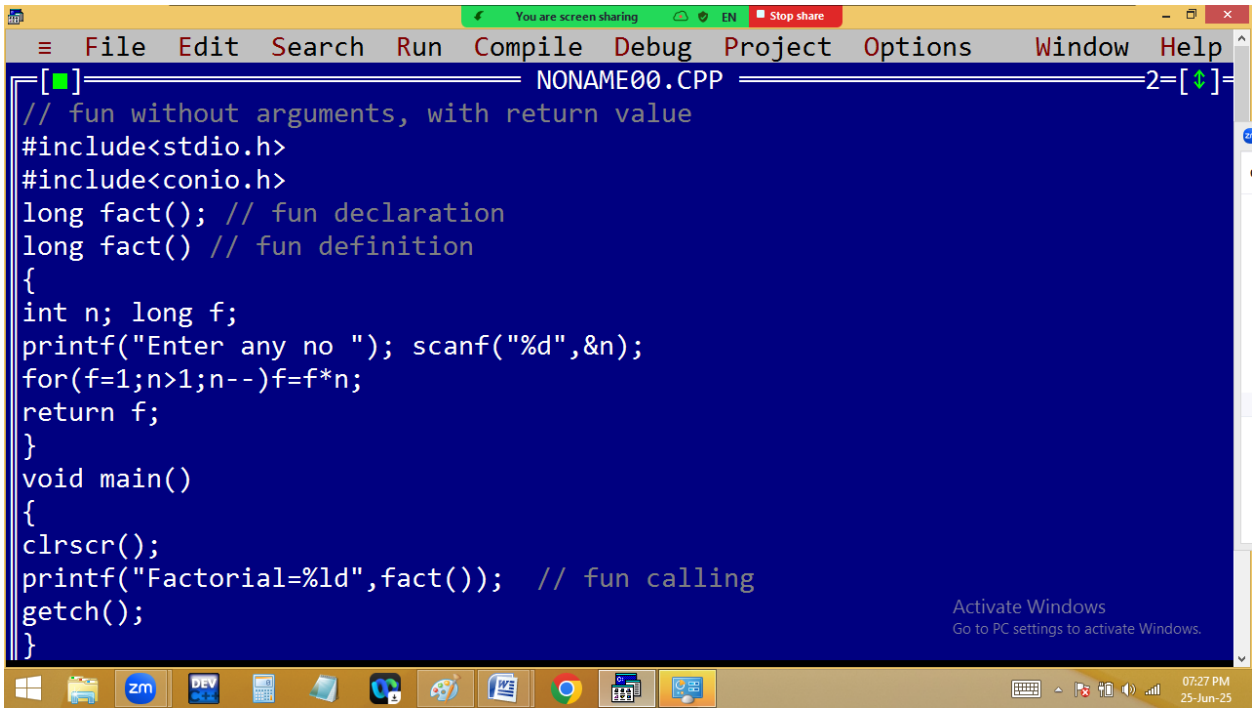


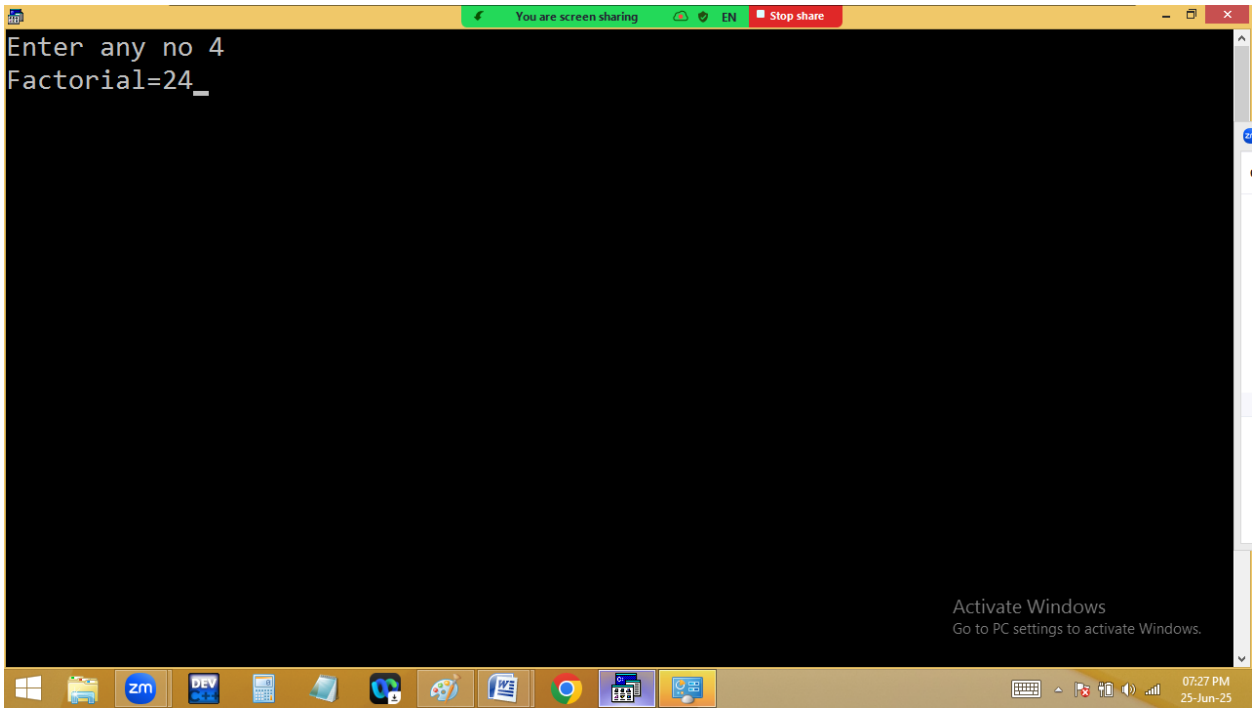
Function without arguments, with return value:

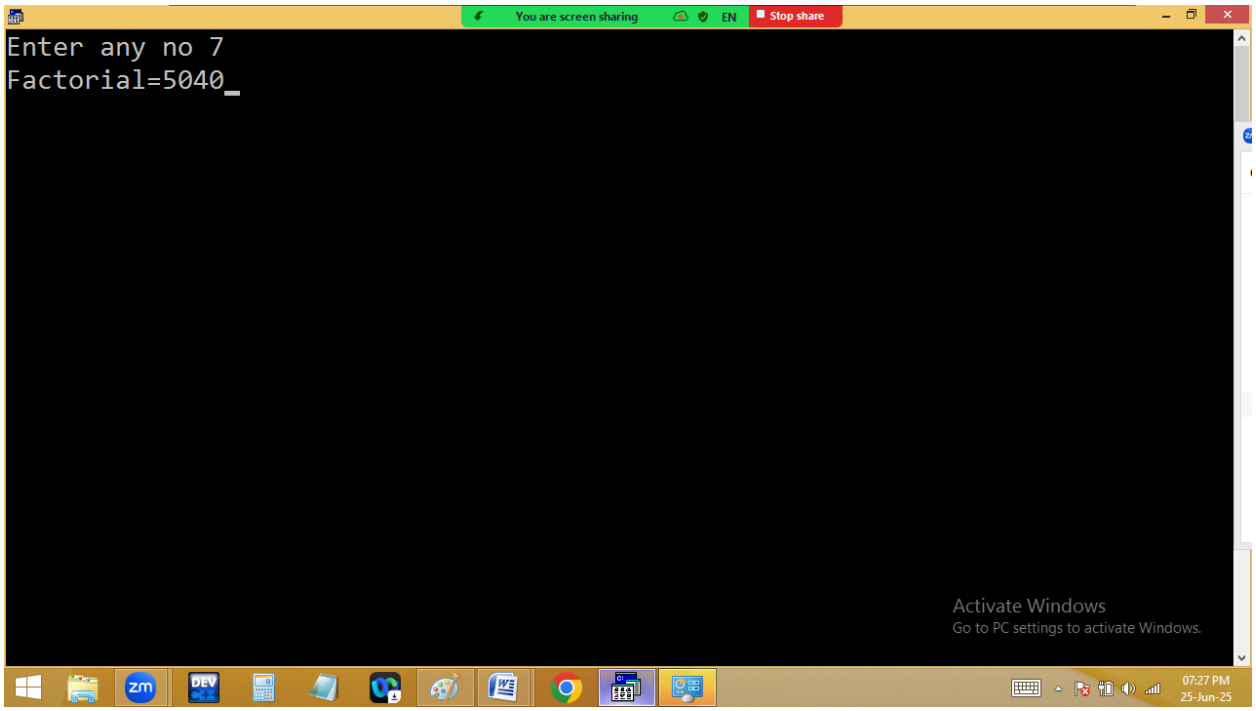

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// fun without arguments, with return value
#include<stdio.h>
#include<conio.h>
int sum(); // fun declaration
void main() // caller
{
int s = sum(); // fun calling, callie
clrscr();
printf("Sum=%d\n",s);
printf("Sum=%d\n",sum()); // fun calling
getch();
}
int sum() // fun definition
{
int a=10, b=20;
return a+b;
}
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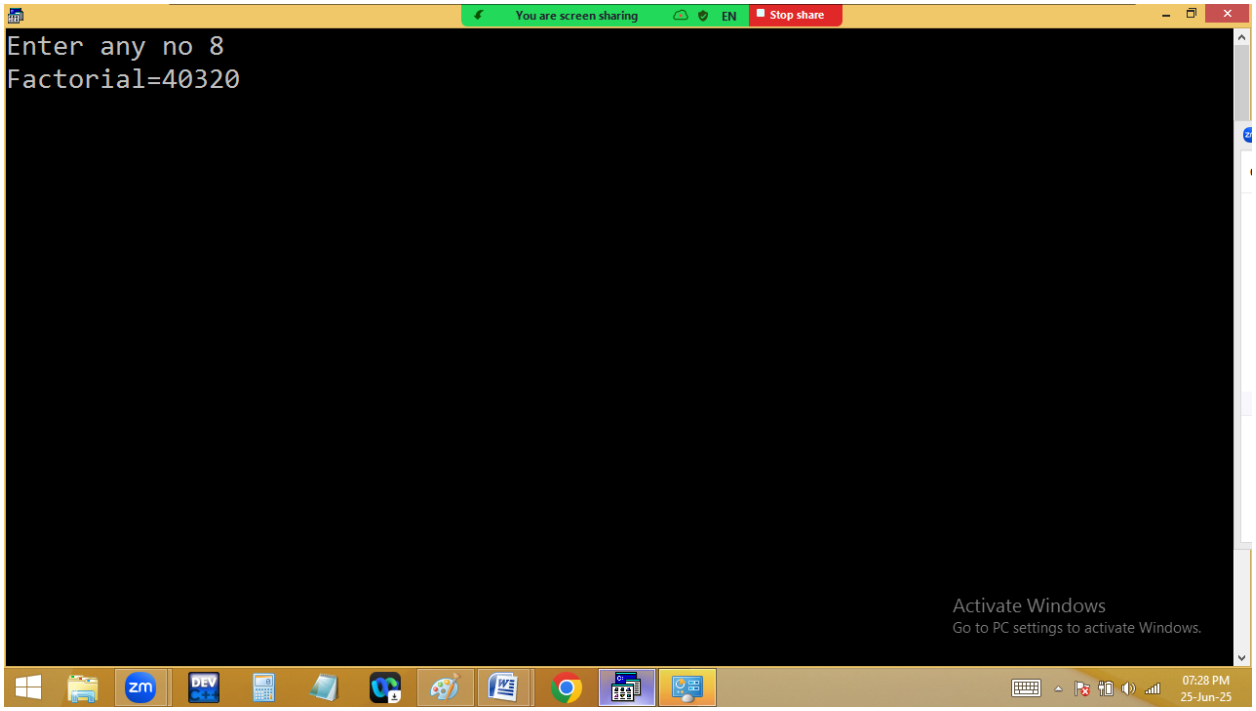

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Sum=30
Sum=30
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Finding factorial using udf?






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Stop share
Enter any no 7
Factorial=5040
Activate Windows
Go to PC settings to activate Windows.
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EN
Stop share
Enter any no 8
Factorial=40320
Activate Windows
Go to PC settings to activate Windows.
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